

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Arithmetic Sequences

Arithmetic sequences move by a constant difference.

ARITHMETIC SEQUENCES · P2 · 05

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

01 DIFF: Common Difference

TRIGGER *arithmetic sequence*

02 TERM: Nth Term

TRIGGER *find u_n*

03 SUM: Arithmetic Sum

TRIGGER *sum of arithmetic sequence*

04 SOLVE: Find Unknowns

TRIGGER *given two terms or a sum*

BANK EVIDENCE Local pages: 17 problem, 30 answer, 6 notes. Website primary entries: 6.

01 DIFF: Common Difference

TRIGGER *arithmetic sequence*

BECOMES Subtract consecutive terms.

FIRST LINE TO WRITE

$$d = u_{n+1} - u_n$$

SIMPLEST STRATEGY

- 1 Detect consecutive terms.
- 2 Identify direction.
- 3 Find the difference.
- 4 Form the next terms.

WORKED MODEL

| Sequence 7, 11, 15 has $d = 4$.

HARD VARIANTS

- 1 If terms are not consecutive, divide the total change by the number of steps.
- 2 A decreasing sequence has negative d .
- 3 Check whether the sequence is arithmetic before using formulas.

— BOTTOM LINE

Arithmetic means add the same amount each time.

02 TERM: Nth Term

TRIGGER *find u_n*

BECOMES Use first term plus $n - 1$ differences.

FIRST LINE TO WRITE

$$u_n = a + (n - 1)d$$

SIMPLEST STRATEGY

- 1 Take first term a .
- 2 Evaluate difference d .
- 3 Replace n .
- 4 Make the requested term or equation.

WORKED MODEL

| If $a = 5, d = 3$, then $u_n = 5 + 3(n - 1)$.

HARD VARIANTS

- 1 For missing term number, set u_n equal to the given value and solve for n .
- 2 Keep n as a positive integer.
- 3 If terms are described in context, define a and d clearly.

— BOTTOM LINE

The first term is not d ; it is a .

03 SUM: Arithmetic Sum

TRIGGER *sum of arithmetic sequence*

BECOMES Use average of first and last times number of terms.

FIRST LINE TO WRITE

$$S_n = \frac{n}{2}(a + l)$$

SIMPLEST STRATEGY

- 1 Secure n .
- 2 Use first and last.
- 3 Multiply average by count.

WORKED MODEL

$$S_{10} = \frac{10}{2}(3 + 30) = 165.$$

HARD VARIANTS

- 1 If last term is unknown, use $l = a + (n - 1)d$ first.
- 2 If the sum is given, form a quadratic in n .
- 3 Reject non-integer or negative term counts.

— BOTTOM LINE

Arithmetic sum is average term times number of terms.

04 SOLVE: Find Unknowns

TRIGGER *given two terms or a sum*

BECOMES Turn the sequence facts into equations.

FIRST LINE TO WRITE

$$u_p = a + (p - 1)d$$

SIMPLEST STRATEGY

- 1 Set up term equations.
- 2 Organise unknowns.
- 3 Linear solve for a, d .
- 4 Verify with given term.
- 5 End with requested value.

WORKED MODEL

$$| \quad u_3 = 10, u_8 = 25 \Rightarrow a + 2d = 10, a + 7d = 25.$$

HARD VARIANTS

- 1 With sums, equations may be quadratic in n .
- 2 Use simultaneous equations when two term positions are given.
- 3 Check the final sequence matches every given fact.

— BOTTOM LINE

Two sequence facts usually find a and d .

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
arithmetic sequence	$d = u_{n+1} - u_n$
find u_n	$u_n = a + (n - 1)d$
sum of arithmetic sequence	$S_n = \frac{n}{2}(a + l)$
given two terms or a sum	$u_p = a + (p - 1)d$